Geometric Hidden Variables for Superdeterminism: Jet Bundle Modes as the Missing Substrate

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Abstract

Superdeterminism restores locality and determinism to quantum mechanics by postulating hidden variables correlated with measurement settings (Hossenfelder & Palmer, 2019). Harmonic geometry establishes an infinite tower of geometric dimensions above any metric manifold at zero algorithmic cost (Bernstein, 2026n). This paper argues that the two results identify the same structure: the hidden variables of superdeterminism are higher-dimensional geometric degrees of freedom in the jet bundle tower. The identification resolves the structural gap in the superdeterministic program and demystifies the conspiracy objection as geometric identity.

Keywords: superdeterminism, harmonic geometry, jet bundle, hidden variables, geometric overtones, Bell correlations, Simplicity Dominance Principle

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1. The Gap in Superdeterminism

Superdeterminism solves the measurement problem. It preserves locality. It requires no branching worlds and no collapse. Its single cost is violating statistical independence: measurement settings must correlate with hidden variables through shared causal ancestry in the initial conditions (Bell, 1964; Hossenfelder, 2020b).

The mainstream objection is fine-tuning: the correlations seem implausibly specific. Bernstein (2026b) dissolves this objection by showing that under algorithmic probability, initial conditions generated by simple rules are measure-favored regardless of their specificity. Simple rules can produce complex correlations at low descriptive cost.

But dissolving the fine-tuning objection leaves a structural question unanswered: what *are* the hidden variables? The superdeterministic program postulates their existence without specifying their physical character. Palmer (2024) proposes fractal invariant sets on which physical states are constrained to lie. Ciepielewski, Okon & Sudarsky (2023) demonstrate dynamical attractor mechanisms that produce the necessary correlations. Neither identifies the geometric substrate.

This paper proposes that the hidden variables are higher-dimensional geometric modes in the jet bundle tower of Bernstein (2026n).

2. The Geometric Tower

The argument from harmonic geometry (Bernstein, 2026n) establishes that an infinite tower of geometric structures exists above any manifold with a metric:

1. A 4D manifold with a metric has a metric bundle: 4 spacetime coordinates + 10 independent metric components = 14D. This is the observation behind Weinstein's Geometric Unity (2021).
2. The first jet bundle above the metric bundle (the space of all first derivatives of the metric) is mathematically necessary.
3. The second jet bundle (derivatives of derivatives) is mathematically necessary.
4. The tower continues without bound.

Each level adds zero algorithmic complexity. Each follows from the one below by the definition of a derivative. The tower exists because calculus exists.

The physics is in the amplitude profile. Like modes on a vibrating string, the fundamental (metric bundle) dominates observable physics. Higher modes contribute progressively smaller corrections. The Standard Model is the timbre of the fundamental. Dark matter may be the first overtone: a conjectural identification developed in Bernstein (2026n).

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3. Identification

We propose the identification:

> The hidden variables of superdeterminism are the higher-dimensional geometric degrees of freedom in the jet bundle tower.

The identification is motivated by three structural parallels:

Locality. Superdeterminism requires the hidden variables to be local: no action at a distance. Higher geometric modes are defined locally at each point of the base manifold. The jet bundle at a point depends only on the metric and its derivatives at that point. The tower preserves locality by construction.

Determinism. Superdeterminism requires deterministic evolution. The geometric tower does not add independent stochastic degrees of freedom; once the base metric dynamics are fixed, jet components are fixed as derived quantities. There is no stochastic element at any level.

Correlation with measurements. Superdeterminism requires that hidden variables correlate with measurement settings. In the geometric picture, a measurement apparatus is a macroscopic configuration of the metric and its derivatives. The higher-dimensional degrees of freedom at the apparatus location are, by definition, the derivatives of the metric configuration that constitutes the apparatus. The correlation is not conspiratorial. It is geometric identity: the apparatus *is* a configuration of the same tower whose higher modes are the hidden variables.

This dissolves the conspiracy objection from a new direction. The hidden variables correlate with measurement settings because both are aspects of the same geometric structure at different levels of the tower. Asking why they correlate is like asking why a guitar string's fundamental correlates with its overtones. They are the same string.

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4. Entanglement as Low-Order Shadow

Bell experiments reveal correlations between spatially separated measurements that violate classical bounds. Under standard quantum mechanics, these require nonlocality or Many-Worlds branching. Under superdeterminism, they require hidden-variable correlations.

Under the geometric identification, the mechanism is explicit. Two particles prepared in an entangled state share a common origin in the base metric. Their jet bundles carry the imprint of this shared origin to all derivative orders. When the particles separate in 3+1 dimensions, the first few derivatives diverge; they look like independent systems. But the higher derivatives retain the shared structure.

Bell correlations are the projection of this higher-order connection onto low-order measurements. We observe violations of classical bounds because classical bounds assume independence at all orders. The particles are independent at low order and connected at high order. The violation is the shadow of the connection. This is a qualitative reinterpretation, not yet a quantitative model; deriving the specific correlation statistics from the amplitude profile of the tower is an open problem.

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5. Further Implications

For an analytic metric, the infinite jet bundle at any single point contains all derivatives at that point. By Taylor's theorem, all derivatives at one point reconstruct the entire function everywhere. Mathematically, every point encodes the full spacetime geometry. This result sits at the top of a hierarchy: Turing encoding maps any dimension to one, preserving data but not physics; the holographic principle (Bousso, 2002; 't Hooft, 1993; Susskind, 1995) maps a volume to its boundary, preserving physics within entropy bounds; jet bundle reconstruction maps a spacetime to a single point, preserving geometry exactly.

These are three distinct senses of encoding, not a single theorem family. The holographic principle, originally derived from black hole thermodynamics, may be a special case of a fact derivable from calculus. This reconstruction assumes analyticity; singularities and non-analytic regimes limit the domain to analytic patches, which is where all observable physics occurs.

Physically, accessing higher derivatives requires coupling to progressively higher-energy modes, and the reconstruction of distant features from local derivatives is generically ill-conditioned. Whether a sufficiently advanced civilization could engineer instruments that read high-order geometric modes without prohibitive backreaction, and what consequences this would have for communication, the Fermi paradox, and the interpretation of decoherence, remains an open and highly speculative question that this paper does not attempt to answer.

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6. Conclusion

Superdeterminism needs hidden variables. Harmonic geometry provides an infinite tower of geometric degrees of freedom at zero algorithmic cost. The identification resolves the structural gap in the superdeterministic program, demystifies the conspiracy objection as geometric identity, and offers a geometric reinterpretation of entanglement as a low-order manifestation of higher-order geometric connection.

The contribution is narrow and specific: to give the hidden variables a name and an address. They live in the jet bundle. They are the derivatives of the metric. They correlate with measurements because measurements are configurations of the same metric whose derivatives they are. The conspiracy is geometry. This paper does not derive Born-rule statistics or Tsirelson bounds; it proposes only the geometric substrate on which such a superdeterministic model could be built.

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